Shivam Randive

Integrated M.Sc. Physics, SVNIT Surat, India | Machine learning for materials prediction

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Research Profile

My research focuses on discovering new alloys in the Heusler family and predicting their properties for thermoelectric applications, with machine-learning predictions tested against the ground truth they replace. One thread runs through my work. At the S. N. Bose Centre I showed that graph neural networks trained on narrow data fail on unseen materials. At SVNIT this grew into two first-author manuscripts with my supervisor, Dr. Himanshu Pandey: an ML screen of 106,235 Heusler structures judged by 3,625 first-principles calculations [P1], and a correction that makes ML predictions of thermal conductivity match experiment [P2]. At NIMS, Japan, I am now extending this to new alkali- and alkaline-earth-metal Heusler alloys, predicting their Seebeck coefficient and lattice thermal conductivity with transfer-learned interatomic potentials. **Interests:** graph neural networks, interatomic potentials, uncertainty-aware screening, generative design.

Education

Integrated M.Sc. (5-year BS–MS) in Physics — **2022 – 2027 (expected)**
Sardar Vallabhbhai National Institute of Technology (SVNIT) — *Surat, India*
CGPA **8.30 / 10** (top 20% of class)
Thesis: ML for materials discovery and property prediction in Heusler thermoelectrics

Semester Exchange Student — **Upcoming (Jan 2027)**
Indian Institute of Technology (IIT) Delhi — *New Delhi, India*

Publications

First-author manuscripts, both submitted to journals — * corresponding author

[P1] **S. Randive**, R. Dixit, H. Pandey. **High-throughput screening of Heusler alloys for thermoelectrics combining machine-learned densities of states and first-principles transport.** Submitted (2026). Preprint: arXiv:2610.18179.

[P2] **S. Randive***, O. Randive, H. Lungaria, H. Pandey*. **Predicting the experimental lattice thermal conductivity of half-Heusler compounds by machine learning and transfer-function calibration.** Submitted (2026); arXiv preprint forthcoming. Code & data: doi:10.5281/zenodo.23333947

Posters & Talks

Poster: *AI-driven discovery of Heusler alloys using generative models and electronic structure learning*, National Science Day, SVNIT Surat — Feb 2026

Research Internships

Research Intern · Spin Theory Group, Prof. Terumasa Tadano — **Sep – Dec 2026**
National Institute for Materials Science (NIMS) — *Tsukuba, Japan*

New alkali- and alkaline-earth Heusler thermoelectrics with pretrained AI models — ongoing

- *Idea:* build new alkali- and alkaline-earth-metal Heusler compositions combinatorially and rank them by predicted Seebeck coefficient and lattice thermal conductivity, using universal interatomic potentials (eSEN, EquiformerV3) pretrained on millions of DFT calculations.
- *Approach:* keep the pretrained network frozen and train a small model on its inner features, finding which layers, and which design (convolution or attention), best capture transport.

Summer Research Fellow (IASc SRFP) · Prof. Tanusri Saha-Dasgupta — **Jun – Jul 2025**
S. N. Bose National Centre for Basic Sciences — *Kolkata, India*

Do graph neural networks work on materials they have never seen? [report]

- Built a full ALIGNN pipeline and compared a broadly pretrained model with "specialists" trained only on narrow perovskite families.
- *Result:* specialists scored >92% on familiar data but only ≈48% on new double perovskites; the pretrained model kept ≈80%. *Message:* breadth of data beats narrow accuracy.

Summer Research Intern — **Jun – Jul 2024**
Indian Institute of Technology (IIT) Bhilai — *Bhilai, India*

Strain in silicon thin films for stretchable electronics [report]

- COMSOL finite-element model of a silicon film on flexible polyimide; its strain maps fed a collaborator's DFT study of how strain tunes the film's electronic properties.

Research Projects

Semester research projects in the Integrated M.Sc., Department of Physics, SVNIT, supervised by Dr. Himanshu Pandey

Testing the ML shortcut used to find new thermoelectrics [P1] 2026

- *Problem:* thermoelectrics turn waste heat into electricity. Searches for new ones first pick candidates from a cheap property, the density of states (DOS), often predicted by ML, before running costly calculations. No one had checked whether this shortcut picks the right materials.
- *What I did:* tested it on all 106,235 Heusler compounds in a large database. Graph neural networks predicted the DOS where none existed, and I checked the shortlist with full first-principles calculations run through my own automated workflow.
- *Result:* the shortcut is good at choosing what to calculate, needing about **4× fewer expensive calculations** per good material, and it found **80 promising candidates**, 61 never measured. But ML spectra often miss whether a material has a band gap, even when accuracy scores look excellent, so a full calculation must confirm it.

Making ML predictions of thermal conductivity match experiment [P2] 2026

- *Problem:* a good thermoelectric must conduct heat poorly. ML models that predict this learn from computer calculations, but I found that these calculations overestimate the measured value, typically by **about 70%**.
- *What I did:* gathered thermal-conductivity data from databases and published papers into one dataset that records where every value came from (AI models helped read the papers; every value was checked against physics), and built a simple correction that turns a calculated value into the one an experiment would measure.
- *Result:* with the correction, ML predicts measured values with **about 30% less error** than standard approaches, close to how much two laboratories disagree. I also released predictions for new compounds that experiments can test.

Simple physical descriptors as first predictors [report] Jan – Apr 2025

- Support-vector classifier for perovskite formability (≈90% accuracy); reproduced a Gaussian-process model of lattice constants for 137 half-Heusler compounds.

Next: Phono3py thermal conductivity for the [P1] candidates; crystal generation with MatterGen.

Open Data & Code

- **heusler-dos-te-screening** [P1] *(I designed and built it)*: the full screening pipeline (ML models, ranking tools, automated DFT transport workflow) with all 3,625 results and the 80 candidates.
- **Predicted DOS dataset** [P1] *(lead author; Zenodo, release pending)*: ML-predicted DOS for **54,978 Heusler structures** with no DFT spectrum, plus the 30 trained models.
- **half-heusler-kappa-calibration** [P2] *(lead author, archived on Zenodo)*: the 9,495-value thermal-conductivity dataset with the source of every value, the fits, the 10 predictions and the code that rebuilds every figure and number in the paper.

Fellowships & Awards

Fully funded research internship, National Institute for Materials Science (NIMS), Japan 2026
Selected for semester exchange at IIT Delhi; eligible for direct PhD admission 2026
Summer Research Fellowship (SRFP), Indian Academy of Sciences (national, merit-based) 2025
Best Structural Stability and Flight Award (3rd overall), national rocketry, IIT Mandi 2025
Xpecto tech fest; designed and launched a 250 g rocket with my team

Technical Skills

ML methods	Graph neural networks (training, fine-tuning, transfer learning); MLIP transfer learning and fine-tuning (eSEN, EquiformerV3); conformal prediction; SHAP
Classical ML	CatBoost, LightGBM, Gaussian processes, support-vector machines
ML libraries	PyTorch, DGL, fairchem, scikit-learn, NumPy, pandas, Matplotlib
First principles	VASP, Quantum ESPRESSO (incl. spin–orbit), BoltzTraP2, Phonopy/Phono3py
Materials data	Materials Project, AFLOW, JARVIS, OQMD, Starrydata; LLM literature mining
Computing	Python, Linux, SLURM GPU clusters, Git, MATLAB, COMSOL Multiphysics, LaTeX

Mentoring

Mentored two junior students, O. Randive and H. Lungaria; both are co-authors on [P2].